CROP YIELD PREDICTION USING REGRESSION FOR TAMILNADU AGRICULTURE



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ABSTRACT

OVERVIEW

Predicts crop yield in Tamil Nadu using machine learning regression based on rainfall, temperature, soil conditions and cultivated area.

EXISTING WORK

Linear Regression, Decision Tree, Random Forest, SVR and ANN are used for crop yield prediction. Some models may suffer from overfitting and limited interpretability.

RESEARCH GAP

Limited district-wise agricultural data, changing climatic conditions, insufficient regional analysis and difficulities in explaining prediction results.

NOVELTY

Tamil Nadu agricultural data analysis, district-wise crop yield prediction, weather-based features and interpretable machine learning results.

PROPOSED

Linear Regression model, agricultural data preprocessing, model evaluation using MAE and R^2 score, and crop yield prediction for selected Tamil Nadu districts.

Introduction:

Crop yield prediction is an important application of machine learning in agriculture. Crop production depends on various factors, including rainfall, temperature, soil conditions, irrigation, and cultivated area. Traditional crop yield estimation methods may not always provide accurate predictions. Machine learning techniques can analyze historical data and identify relationships between agricultural factors and crop production. In this mini project, a regression algorithm is used to predict crop yield from agricultural data related to Tamil Nadu. The system takes relevant agricultural parameters as input and generates an estimated crop yield as output. The main objective is to develop a simple and useful prediction system that supports better agricultural planning.

What we learned from the project

Aspect	What Worked	Challenges
Data Source	Government reports provided reliability	Manual entry was time consuming
Algorithms	Random Forest handled noise well	Linear models were too simple
Features	Rainfall was a great predictor	Soil data was hard to find locally
User Interface	Python scripts were easy to run	Deployment needs more work

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Our development methodology



Technologies Used:

- Programming Language: Python
- Machine Learning Algorithm: Linear Regression
- Libraries: Pandas, NumPy, Scikit-learn, Matplotlib
- Dataset: Historical agricultural production, rainfall, temperature, and cultivated-area data for Tamil Nadu
- Output: Predicted crop yield

Existing System:

1. Crop yield estimation in Tamil Nadu today relies mostly on traditional, manual methods:
2. Farmer experience and guesswork: Farmers estimate yield from past seasons and local knowledge.
3. Crop cutting experiments: Government departments sample small plots and extrapolate to the whole district. This is slow, costly, and labor-intensive.
4. Manual record keeping: Data from agriculture offices (area, production, rainfall) is stored in registers or separate files and is hard to analyze.
5. No integration of factors: Rainfall, temperature, soil type, and fertilizer use are rarely combined in one prediction.
6. Delayed and inaccurate results: Estimates arrive after harvest, so they are of little use for planning, pricing, or crop insurance.
7. Drawbacks: Low accuracy, time-consuming, subjective, not scalable, and no support for decisions such as crop selection or resource planning.

PROPOSEDSYSTEM:

1. A REGRESSION-BASED MACHINE LEARNING SYSTEM THAT PREDICTS CROP YIELD FROM HISTORICAL AGRICULTURAL AND CLIMATE DATA FOR TAMIL NADU.
2. HOW IT WORKS:
3. DATA COLLECTION: DISTRICT-WISE DATA ON CROP TYPE, AREA, RAINFALL, TEMPERATURE, SOIL TYPE, SEASON, AND FERTILIZER USE.
4. PREPROCESSING: HANDLE MISSING VALUES, ENCODE CATEGORICAL VARIABLES (DISTRICT, CROP, SEASON), REMOVE OUTLIERS, AND NORMALIZE FEATURES.
5. MODEL TRAINING: APPLY REGRESSION ALGORITHMS SUCH AS LINEAR REGRESSION, RANDOM FOREST REGRESSION, DECISION TREE REGRESSION, AND OPTIONALLY XGBOOST.
6. EVALUATION: COMPARE MODELS USING R^2, MAE, MSE, AND RMSE, THEN CHOOSE THE BEST PERFORMER.
7. PREDICTION: THE USER ENTERS DISTRICT, CROP, SEASON, AND WEATHER INPUTS, AND THE SYSTEM OUTPUTS THE EXPECTED YIELD.
8. INTERFACE: A SIMPLE WEB OR APP FRONT END (FLASK OR STREAMLIT) THAT FARMERS AND OFFICERS CAN USE.

Advantages:

- Higher accuracy than manual estimation
- Early prediction before harvest
- Combines multiple factors in one model
- Helps farmers choose crops and plan resources
- Supports government policy, pricing, and insurance decisions
- Scalable to new districts and crops

Disadvantages :

- Data dependency: Accuracy depends on the quality, quantity, and completeness of historical data. Missing or outdated district records weaken predictions.
- Unpredictable events: Floods, droughts, pests, and disease outbreaks are hard for regression to capture.
- Model limitations: Simple regression assumes mostly linear relationships and may miss complex interactions between weather, soil, and crop type.
- Limited factors: Features like soil nutrients, irrigation method, seed variety, and farming practices may not be available, which reduces accuracy.

Conclusion:

Traditional crop yield estimation in Tamil Nadu is slow, manual, and often inaccurate, which makes planning difficult for farmers and officials. This project uses regression-based machine learning to predict crop yield from historical data such as rainfall, temperature, soil type, season, crop type, area, and fertilizer use. By comparing models like Linear Regression and Random Forest using R^2, MAE, and RMSE, the best model is selected to give earlier and more accurate predictions than manual methods. The system helps farmers choose suitable crops and plan resources, and it supports government decisions on policy, pricing, and insurance. Its accuracy depends on data quality, and it cannot fully predict events like floods, droughts, or pests. In future, real-time weather data, satellite imagery, deep learning models, and a Tamil-language mobile app can make it more accurate and easier for farmers to use.